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Hamstring exercises for track and field athletes: injury and exercise biomechanics, and possible implications for exercise selection and primary prevention

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ABSTRACT

Hamstring strain injuries are the most prevalent muscle injuries in track and field (TF). These injuries often cause prolonged symptoms and a high risk of re-injury. Strengthening of the hamstring muscles has been recommended for injury prevention. The authors review the possible role of eccentric training in TF hamstring injury prevention and introduce exercise classification criteria to guide clinicians in designing strengthening programmes adapted to TF. The principles exposed may serve as a foundation for future development and application of new eccentric programmes to decrease the high incidence of this type of injury in other sports.

INTRODUCTION

Muscle strains of the posterior thigh muscles (PTM), the 'hamstrings', present the highest index of prevalence[1–12] and re-injury rates[12 13] in sport and concretely in track and field (TF),[1 3 12 13] as reported in many publications.

An injury prevention approach for PTM would consider the interconnected, multidirectional and synergic interaction between various risk factors involved in this injury: core stability, range of motion, architecture, strength, fatigue and neuromuscular control.[14] The value of this article for clinicians is its focus on strength factor in relation to hamstring strains. We review the possible role and effects of eccentric strength training for injury prevention and use that, together with injury biomechanics, as a basis to suggest an eccentric exercise classification criteria applicable to the TF athletes.

THE ROLE OF ECCENTRIC EXERCISES IN TF AND HAMSTRING INJURY PREVENTION

A good understanding of injury mechanism is a prerequisite for any successful preventive intervention. The hamstring muscles act as hip extensors and knee flexors during both stance and swing phase of sprinting, the most common mechanism of injury in TF athletes. It is hypothesised that the hamstrings are susceptible to injury during terminal swing phase.[15–18] This can be explained by three reasons. First, peak hamstring musculotendinous stretch seems to occur during the late swing phase of sprinting before foot contact. Musculotendinous stretch is significantly greater for biceps femoris, probably because of a shorter knee extension moment arm.[18] Second, electromyography data indicate that the hamstrings are active at this same phase of the gait cycle.[19] Third, the hamstrings are undergoing an active lengthening contraction during late swing, producing the potential conditions for a strain injury to occur.[15–18] Even though the hamstring muscles are more likely to sustain strain injuries during the late swing phase than during the late stance, Yu *et al* recently reported that the hamstring muscles undergo a double peak of eccentric contraction, that is, during the late stance phase and during the late swing phase of overground running.[20]

Strength as a risk factor in TF

One of the proposed risk factors for acute hamstring injuries in TF athletes is muscle weakness during concentric and/or eccentric contractions.[21 22] Muscle weakness has been assessed with one of two methods: (1) comparing the peak torque values of the knee extensors (during concentric contraction) with their antagonistic muscle group that is the knee flexors (during concentric or eccentric contraction); and (2) comparing the peak torque values of one leg with the contralateral leg during knee flexion. Both methods have produced conflicting findings in prospective studies. Sugiura *et al* recently reported that eccentric peak torque (functional ratio and side to side) was significantly decreased in six sprinters who sustained an acute hamstring injury over a 12-month period.[21] In contrast, Yeung *et al* found a decreased concentric H/Q ratio only at 180° with no eccentric deficits between eight injured sprinters and the remaining 36 sprinters in the cohort evaluated at preseason.[22] Evidence is still weak due to lack and unpowered prospective studies and therefore, the influence of strength on the risk of hamstring injury in TF requires further investigation.

Eccentric training properties

Eccentric training overloads the muscle to a greater extent and enhances muscle mass, strength and power than concentric training.[23 24] Force of this magnitude (in excess of the maximum isometric force) is only possible during eccentric (vs isometric or concentric) contractions. Eccentric contractions not only produce the highest forces, but also do so at a greatly reduced energy expenditure.[25 26]

Eccentric muscle action plays a great role in the stretch-shortening cycle (SSC) where it precedes concentric action.[27] Specifically, the hamstring muscle tendons undergo eccentric lengthening from

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approximately 45% to 90% of the sprinting gait cycle, and thereafter shorten before foot contact.[15] This SSC results in improved running economy by a significant enhancement of the power output of the subsequent contraction.[27–29] As a result, the hamstrings are undertaking a substantial amount of negative work during late swing phase. Negative musculotendinous work performed by the hamstrings is increased considerably with speed.[15] In this capacity, the hamstrings and their tendons are behaving as springs that cyclically absorb and recover elastic energy before foot contact during the sprinting gait cycle. This function is significantly time dependent and, if not recovered, the energy is lost as heat.[29] Hence, combining both properties (shock absorber and time dependent spring), the hamstring muscles likely function as a shock absorber in series with a spring.[23 29] Chronic exposure to eccentric muscle activity results in an active spring structure(s) adaptation (ie, the muscle stiffens) in addition to the above mentioned load absorption and strength capabilities.[23 24] Biomechanically, muscle stiffness is the ratio of force response that results from and resists mechanical stretch. Therefore, a stiffer muscle could act to protect the stretching muscle from stretch overload damage, and at the same time to enhance the amount of elastic recoil energy available in the SSC.[23–25] However and paradoxically, there is evidence that chronic eccentric training results in increased flexibility of hamstring muscles measured using joint range of motion.[30]

Eccentric exercise may also prevent injury to the muscle-tendon unit by improving the muscle's ability to absorb more energy and increased force before failing.[23 24] The exact mechanism of this adaptation is not defined, but is likely a result of both structural and neural influences. It is apparent, however, that, if the tissue failure force threshold increases and the attenuation of loads are enhanced, a protective effect can occur.

Skeletal muscles have an optimum length for producing peak tension. Muscle strain injuries are thought to occur when activated muscles are lengthened to greater than optimal length.[31–33] The hamstring muscles are actively lengthened during hip flexion and knee extension, which occur simultaneously during the late swing phase in running (ie, as the air borne leg swings forwards). Different authors suggest that, athletes who produce peak torque at shorter length are more prone to injury because part of the hamstring muscle is operating in a risk range of the length tension, as may occur during late swing phase of sprinting.[31 32 34] A recent retrospective study has identified the optimum length as a risk factor for injury. Brockett *et al* measured the optimum lengths in some TF athletes with previously injured hamstrings.[31] One leg served as the experimental leg (ie, previously injured hamstring), and the other leg served as the control leg (ie, uninjured hamstring). The previously injured hamstring produced peak tension at 12.7° less than the uninjured hamstring (ie, shorter optimum length). It has been argued that hamstring injuries can be reduced if this optimum length can be increased through training.[31 32 34] The only form of training that has been shown to consistently increase the optimum length of tension development has been eccentric exercise.[35–39] The addition of sarcomeres in series (sarcomerogenesis) related to a increased fascicle length and an increase in passive tension at longer length has been proposed to explain optimum length changes after eccentric exercise.[30 34] Furthermore, eccentric training has consistently shown to be able to reduce hamstring injury rates.[3 33 40–42] This may explain the success achieved by the Nordic hamstring exercise, the most common exercise used in literature to prevent such injuries.

Eccentric training and hamstring prevention

Since the Nordic hamstring exercise was described in 2001 by Brockett, five intervention studies aimed at reducing the incidence of hamstring injuries have focused on the use of eccentric training.[32 33 40–42] Four of the studies used the Nordic hamstring exercise,[32 33 40 42] and one study used the yo-yo hamstring curl exercise.[41] More than 1000 athletes were monitored in these five studies and each study reported significant reductions in injury rates. Both Nordic hamstring and yo-yo hamstring curl exercises increase the strength[30 36] and the angle at which the hamstring produce their peak torque.[30 35–39]

Thus, optimum length at a given strength may be an important variable for assessing injury risk and monitoring progression of an eccentric-based intervention.

In summary, eccentric training has the properties to increase the size, strength and flexibility of hamstring muscles, change hamstring muscle optimum length and stiffening of the muscle spring that can occur independent of, or in addition to, increases in size and isometric strength of the muscle.[23 24 29 30]

EXERCISE IMPLEMENTATION RATIONALE

Brughelli and Cronin[43] suggested a combination of uni- and bilateral exercises based on the principle of avoiding muscle strength asymmetry and possible overcompensations. This is a clinical recommendation – it has not yet been tested empirically. More recently, core stability concept has been strongly linked with hamstring injuries.[44] Decreased neuromuscular control of the core as a possible cause of injury justifies the increased use of current trunk or core neuromuscular training as a central tenet of rehabilitation treatments and prevention.[44 45] However, core stability use requires further investigation.[14] Combination of multi-joint exercises in addition to only localised exercises will ensure core involvement during different exercises.

Selection criteria in order to classify and programme a complete eccentric exercise programme for primary prevention of hamstring injury

Injury mechanism

As stated above, active lengthening of the hamstring muscles may occur both in the late swing phase (open kinetic chain) and during late stance phase (closed kinetic chain) of sprinting.[15–18 20] This suggests that open and closed kinetic chain exercises should be included in the designed prevention programme. Furthermore, the hamstrings have a key role in the SSC.[27 28] The hamstrings lengthen under load from 45% to 90% of the gait cycle (swing) absorbing imposed mechanical energy,[15] and then shorten under load from late swing through stance to reuse this energy. In this way, they no longer act only as shock absorbers; rather, they perform more like springs.[24 29] Therefore, we strongly advise to use SSC exercises and combine them with isolated eccentric exercises in open or closed kinetic chain in order to replicate hamstring function. A more detailed description of some exercises designed to cover the different injury mechanism suggested are shown in figures 1–4.

Hip or knee dominant

When sprinting, the function of the hamstrings in the late swing phase is that of hip extension concentrically, acting to quickly swing the thigh backwards, while also acting as knee flexors to eccentrically decelerate the forward swing of the lower leg. Therefore, the hamstrings show a dual role

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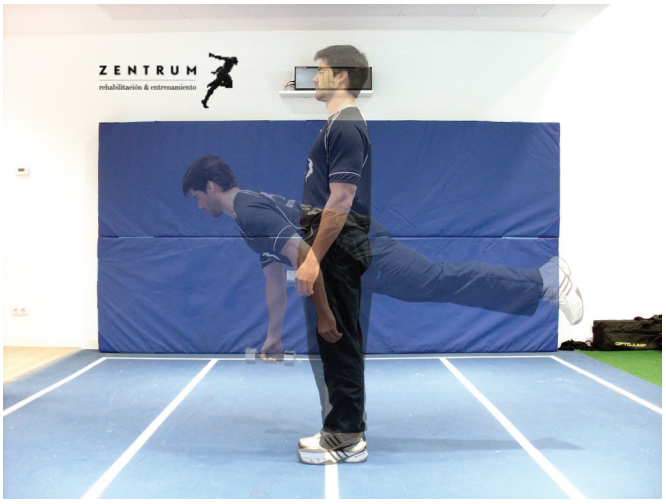

Figure 1 Single leg deadlifts. Example of closed kinetic chain (fixed distal segment) single leg HIP dominant exercise. Hip 'hinge' and lumbopelvic neutral alignment and neuromuscular control is required.

Figure 2 Slide leg curl. Example of combined hip and knee open kinetic chain (mobile distal segment) double leg exercise. Combined hip extension and knee flexion exercise. Eccentric knee flexion is done during descending phase and concentric knee flexion concomitantly with hip extension during the ascending phase.

Figure 3 Hamstring catapult. Example of open kinetic chain, single leg HIP dominant exercise. Concentric hip extension is done during descending phase and eccentric hip extension to counteract hip flexion is done during recovering phase. Core muscle activation must be emphasised during the exercise.

Figure 4 Sprinter eccentric leg curl. Example of open kinetic chain single leg knee dominant exercise. Concentric knee flexion during pulling and eccentric knee flexion to counteract cable imposed forces is done. Stability of the hip at the weight bearing leg and neuromuscular control of the core in order to counteract the torsional forces created at the trunk are required.

for which they have to be prepared. Deficits in eccentric knee extension have been proposed as an injury predictor. Both Nordic hamstring and eccentric leg curl (yo-yo) exercise appeared as the most successful and safe choice to reverse this deficiency. However, during swing phase the moment arm and internal moments at the hip are double that at the knee,[46] and the fascicle length of the hamstring muscles (biceps femoral – BF mainly) are more sensitive to hip position.[47 48] Therefore, a comprehensive prevention programme may include both hip and knee roles of the hamstrings to reproduce their function during sprinting. A more detailed description of some of these examples are shown in figures 1 and 4–6.

Location

Since hamstring injuries may occur either proximally or distally from the insertion[49] in TF sprinters, both locations should be trained with exercises that actively lengthen the hamstrings with either hip flexion, knee extension or a combination of both.[43] Moreover, the non-uniform activation of the different components of the hamstring muscle group during different strengthening exercises has recently been demonstrated,[50] suggesting that we should consider targeting different parts of the hamstring with different exercises. For example, lunge exercise activate the proximal adductor and the proximal biceps femoris[51] in contrast to Nordic hamstring exercises and eccentric leg curls that activate the distal portion of the short head of biceps femoris (unpublished data) and the proximal semitendinosus (ST).[50]

Targeted muscle

Because hamstring strains affect different hamstring muscles, appropriate exercise selection is crucial to target the desired muscle(s). In TF athletes, the long head of the biceps femoris

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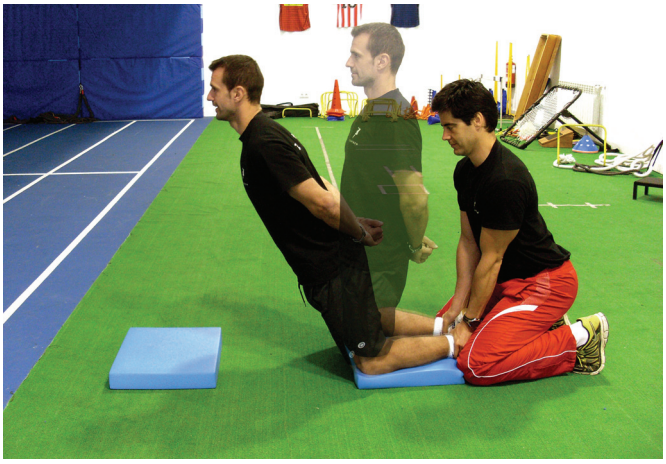

Figure 5 Nordic hamstring. Example of open kinetic chain bilateral knee dominant exercise. Hamstring muscles maximally contract against the knee-extending action as lowering throughout the range of motion while maintaining perfect neutral alignment between trunk and hip joints.

Figure 6 Barbell leg curl. Example of closed kinetic chain (fixed distal segment) double leg HIP dominant exercise. Concentric knee flexion during ascending and eccentric knee flexion during descending is created with the only movement of hip. Barbell grip add posterior trunk muscle involvement.

muscle constituted the primary injury site, and in our opinion needs special consideration. Therefore, when the goal of a therapeutic intervention is to specifically strengthen the ST, adductor magnus, semimembranosus (SM) and/or BF, a progressive resistive training programme that incorporates eccentric leg curls, lunges and deadlifts is indicated, as these exercises selectively and effectively activate the aforementioned muscles.[50–52]

Think more about length than about strength and contraction velocity

Muscle strain injuries are thought to occur when activated muscles are lengthened to greater than optimal lengths.[31–33] Lieber *et al* demonstrated that it is not force per se that causes the muscle damage after eccentric contraction but the magnitude of active strain (strain during active lengthening).[53] This finding is supported by Schache *et al*, who showed that, while

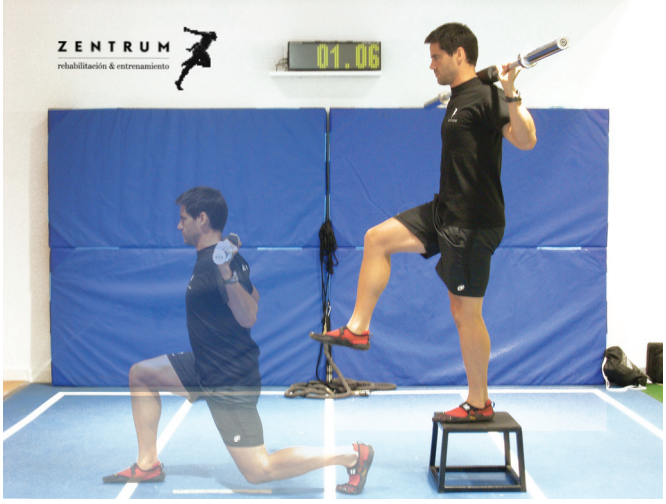

Figure 7 Loaded lunge box drops. Example of progression in strain. Single leg closed kinetic chain hip dominant exercise. Drop jump from a height is done and therefore strain is achieved at the hamstrings due to increases in external hip flexion moments counteracted by internal hip extension moments eccentrically.

peak hamstrings' force during terminal swing did not decrease in the next step postinjury, both peak hamstrings' length and negative work during terminal swing were considerably reduced.[54] Recently, Barroso *et al* indicated that the contraction velocity does not influence strength loss, muscle damage or the repeated-bout effect, supporting the concept that eccentric contraction is not influenced by velocity.[55] To increase the optimum length and strain, eccentric exercises which actively lengthen the hamstrings with either hip flexion, knee extension or a combination of both have been proposed.[43]

Exercise progression must proceed more in relation to length and strain parameters than to strength intensity and contraction velocity. Progressively increasing the length at which muscle groups are trained or modifying ground reaction forces moment arm in relation to the joint may also help to minimise initial muscle damage while gradually improving outer range muscle strength and protect against future injury risk. A more detailed description of how to progress from a simple lunge exercise in order to increase strain is shown in figure 7.

Training parameters

There are currently no published studies comparing different eccentric exercise training protocols to prevent PTM. Volume and loading are often to the discretion of the strength and conditioning coach or rehabilitation clinician. However, even though we consider length more important than strength, we strongly advise that training parameters should follow the common guidelines applied to any strength or rehabilitation programme.[56] The optimal intensity of eccentric training programmes is not yet clear. Some authors recommend that intensity should be high to provide the stimulus necessary to produce further adaptations,[34 57 58] others have found that the protective effect of eccentric training may be observed even using low resistance.[59 60] If strength gains are required to address a strength deficit, eccentric actions should be overloaded between 20% and 80% beyond the maximal isometric strength.[56] However, the volume and intensity of eccentric training programmes should be gradually progressed to minimise the effect of exercise-induced muscle

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What is already known on this topic

- Hamstring strain injuries are the most prevalent muscle injuries in track and field (TF).
- Eccentric training has consistently shown to be able to reduce hamstring injury rates.

What this study adds

- Introduce an exercise classification criteria to guide clinicians in designing prevention strengthening programmes adapted to TF
- Establish principles and factors for analysis of any other sport in order to structure an eccentric intervention.

damage and to provide the stimulus necessary to produce ongoing adaptations.[54 61–63]

In terms of fatigue, several studies have reported different energetic cost for eccentric contractions, with less metabolic demands in terms of oxygen consumption, ATP turnover or ammonia and lactate production compared with concentric exercise.[64–66] However, given the important strength and neuromuscular impairments present immediately after eccentric exercise,[67 68] we advocate the use of eccentric exercise at the end of training sessions.

We recommend the use of highest intensities only with strength athletes already familiarised with this type of training. It would not be recommended to apply immediately an advanced programme of new exercises to an elite athlete without considering the adaptation process that the human body undergoes at new loads and new muscular spectrum performance. When planning the annual training calendar, consideration should be given to avoid high-intensity and long-length eccentric exercise during important competition phases, as the side effects of transient muscle soreness and strength deficits may impair performance. These training recommendations are derived from basic science and clinical experience and as we said above, randomised controlled trial studies comparing different eccentric exercise training programmes effects on hamstring injury prevention and on performance are needed in order to build solid knowledge in this area.

CONCLUSION

Hamstring strain prevention should include specific muscle strengthening. Recent evidence suggests that the protocols of eccentric training can reduce hamstring injuries. To guide clinicians in designing effective and complete strengthening programmes, the basic effects and selection principles of eccentric training are exposed, using TF as an example. However, the same principles and factors can be analysed and applied when structuring an eccentric intervention for any other sport.

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